The four rules of powers of 10

1. Multiplying: ADD the exponents

$$10^a \cdot 10^b = 10^{a+b}$$

$10^3 \cdot 10^4 = 10^{3+4} = \mathbf{10^7}$ · $10^5 \cdot 10^{-2} = 10^{5+(-2)} = \mathbf{10^3}$

With numbers: multiply the numbers and add the exponents.

2. Dividing: SUBTRACT the exponents

$$10^a / 10^b = 10^{a-b}$$

$10^7 / 10^3 = 10^{7-3} = \mathbf{10^4}$ · $10^2 / 10^{-3} = 10^{2-(-3)} = \mathbf{10^5}$

top exponent minus bottom exponent

3. Moving up or down: CHANGE the sign

$$\frac{1}{10^b} = 10^{-b}$$

$\frac{1}{10^3} = \mathbf{10^{-3}}$ · $\frac{5}{10^{-2}} = \mathbf{5 \cdot 10^2}$

A power of 10 can go up or down the fraction line if its exponent changes sign.

4. Power of a power: MULTIPLY the exponents

$$(10^a)^b = 10^{a \cdot b}$$

$(10^2)^3 = 10^{2 \cdot 3} = \mathbf{10^6}$ · $(10^{-3})^2 = 10^{(-3) \cdot 2} = \mathbf{10^{-6}}$

Used to change units of surface (^{2}) and volume (^{3}).